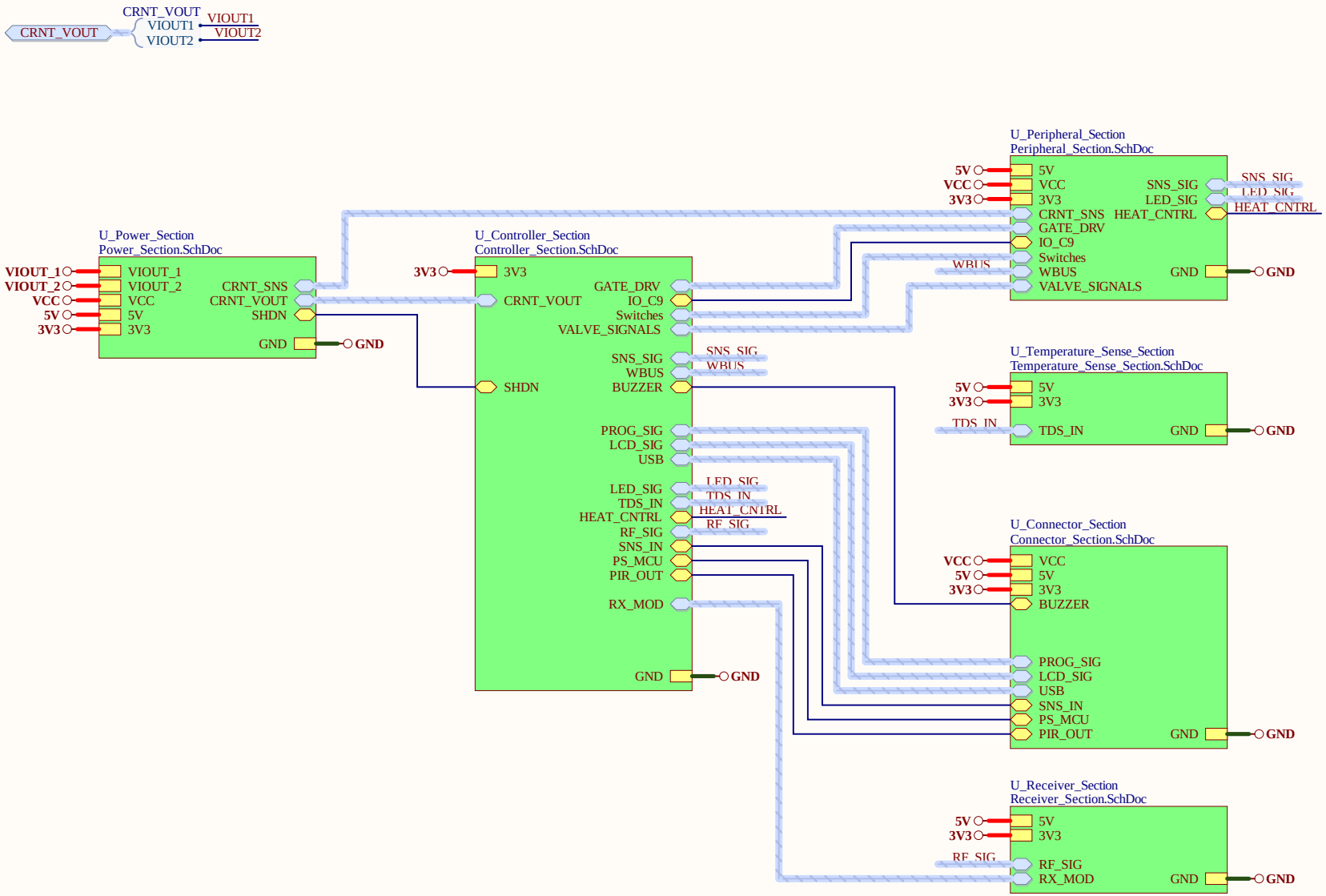
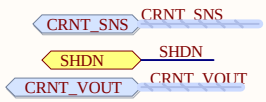
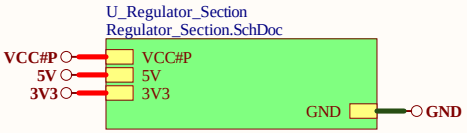
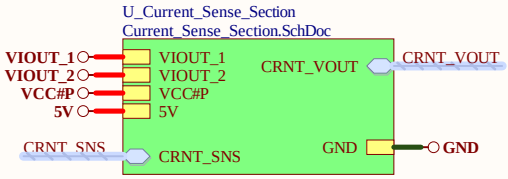
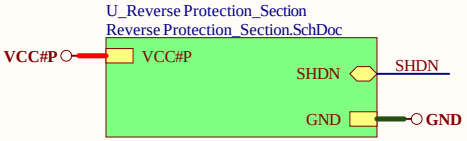
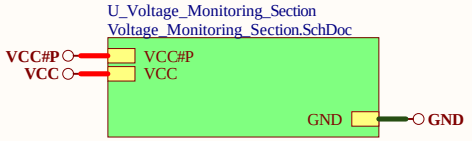

CRNT_VOUT
CRNT_VOUT
VIOUT1
VIOUT2
VIOUT1
VIOUT2
U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VIOUT_1
VIOUT_2
VCC
5V
3V3
CRNT_SNS
CRNT_VOUT
SHDN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
CRNT_VOUT
SHDN
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
SNS_SIG
WBUS
BUZZER
PROG_SIG
LCD_SIG
USB
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
SNS_IN
PS_MCU
PIR_OUT
RX_MOD
GND
GND
SNS_SIG
WBUS
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
U_Peripheral_Section
Peripheral_Section.SchDoc
5V
VCC
3V3
CRNT_SNS
GATE_DRV
IO_C9
Switches
WBUS
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
GND
GND
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
5V
3V3
TDS_IN
TDS_IN
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VCC
5V
3V3
BUZZER
PROG_SIG
LCD_SIG
USB
SNS_IN
PS_MCU
PIR_OUT
GND
GND
U_Receiver_Section
Receiver_Section.SchDoc
5V
3V3
RF_SIG
RF_SIG
RX_MOD
GND
GND

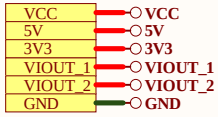

CRNT_SNS
CRNT_SNS
SHDN
SHDN
CRNT_VOUT
CRNT_VOUT


U_Regulator_Section
Regulator_Section.SchDoc
VCC#P
5V
3V3
VCC#P
5V
3V3
GND
GND


U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_VOUT
CRNT_VOUT
CRNT_SNS
CRNT_SNS
GND
GND

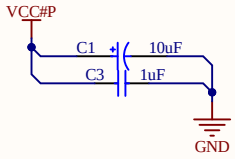

U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC#P
SHDN
SHDN
GND
GND


U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC
VCC#P
VCC
GND
GND

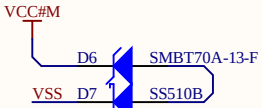

VCC
5V
3V3
VIOUT_1
VIOUT_2
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND

SHDN SHDN

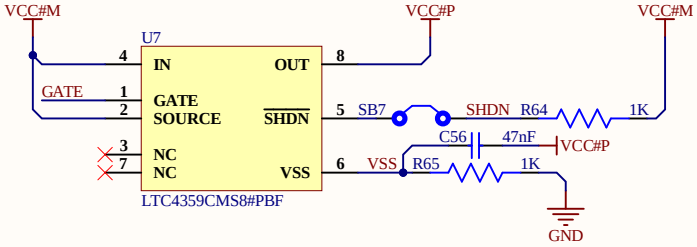
DECOUPLING CAPACITOR



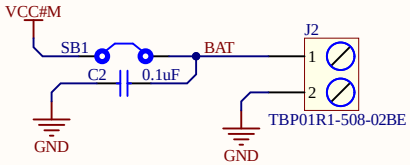
INPUT DIODES



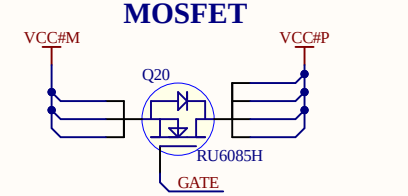
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT

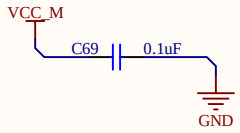


MOSFET

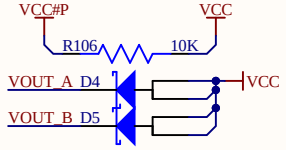


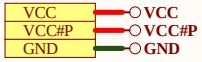
VCC#P VCC#P
GND GND

DECOUPLING CAPACITOR

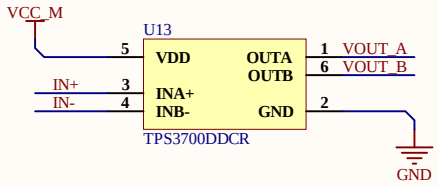


AND GATE CIRCUITRY

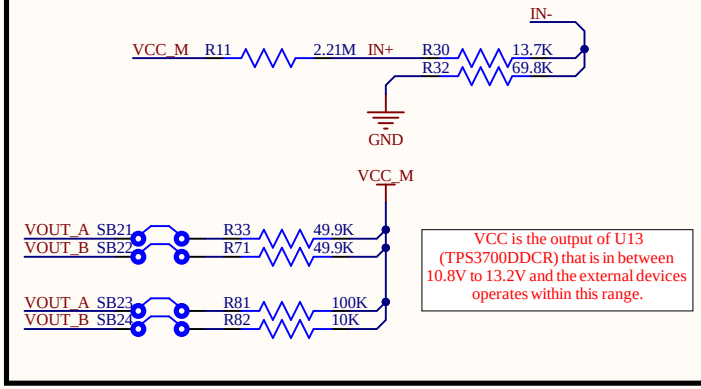




UNDER AND OVER VOLTAGE PROTECTION IC



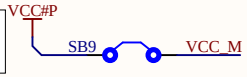
SUPPORTING CIRCUITRY



VCC is the output of U13 (TPS3700DDCR) that is in between 10.8V to 13.2V and the external devices operates within this range.

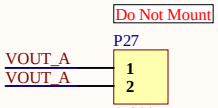
VOLTAGE TO MONITOR

Monitors Main 12V Input. Need 0R Resistor or Shorting Jumper to provide connections to Solder Bridge (SB9).



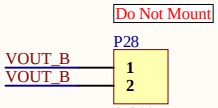
UNDER VOLTAGE OUTPUT

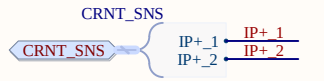
Under Voltage Open Drain Output. 100K Pullup is recommended.



OVER VOLTAGE OUTPUT

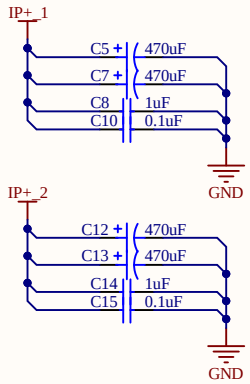
Over Voltage Open Drain Output. 10K Pullup is recommended.



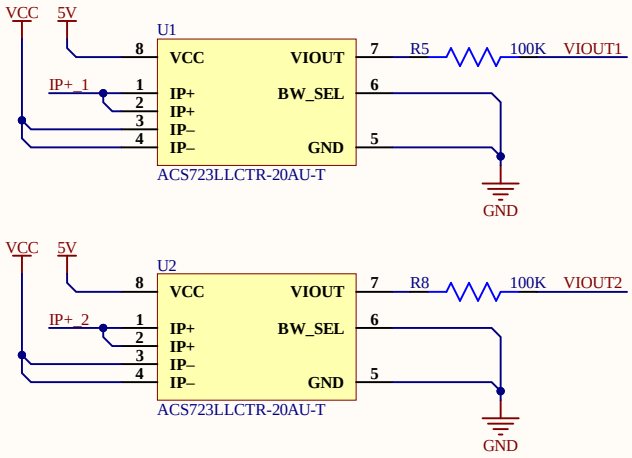




DECOUPLING CAPACITOR

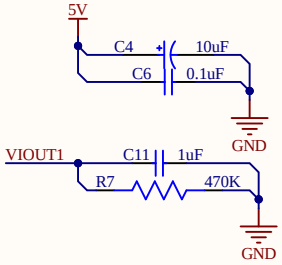


CURRENT SENSE IC

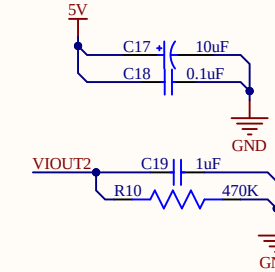


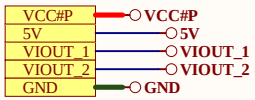
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

VOUT1 SUPPORTING CIRCUIT

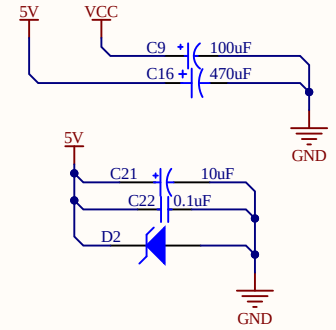


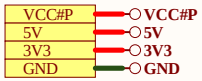
VOUT2 SUPPORTING CIRCUIT



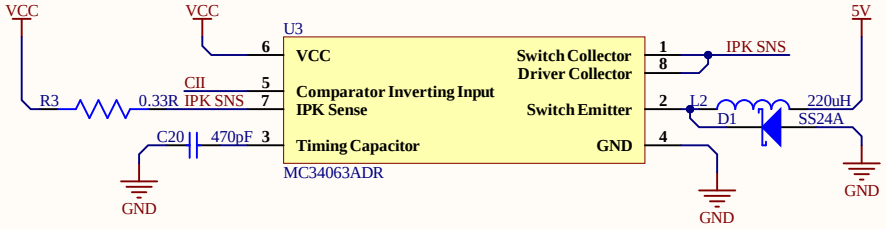


DECOUPLING CAPACITOR

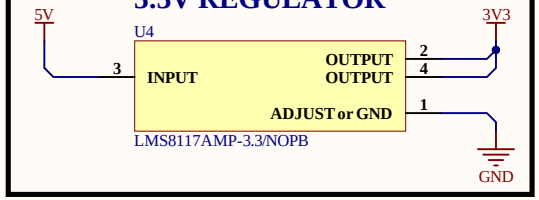




5V BUCK REGULATOR

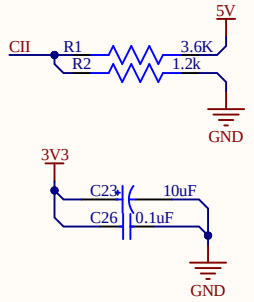


3.3V REGULATOR

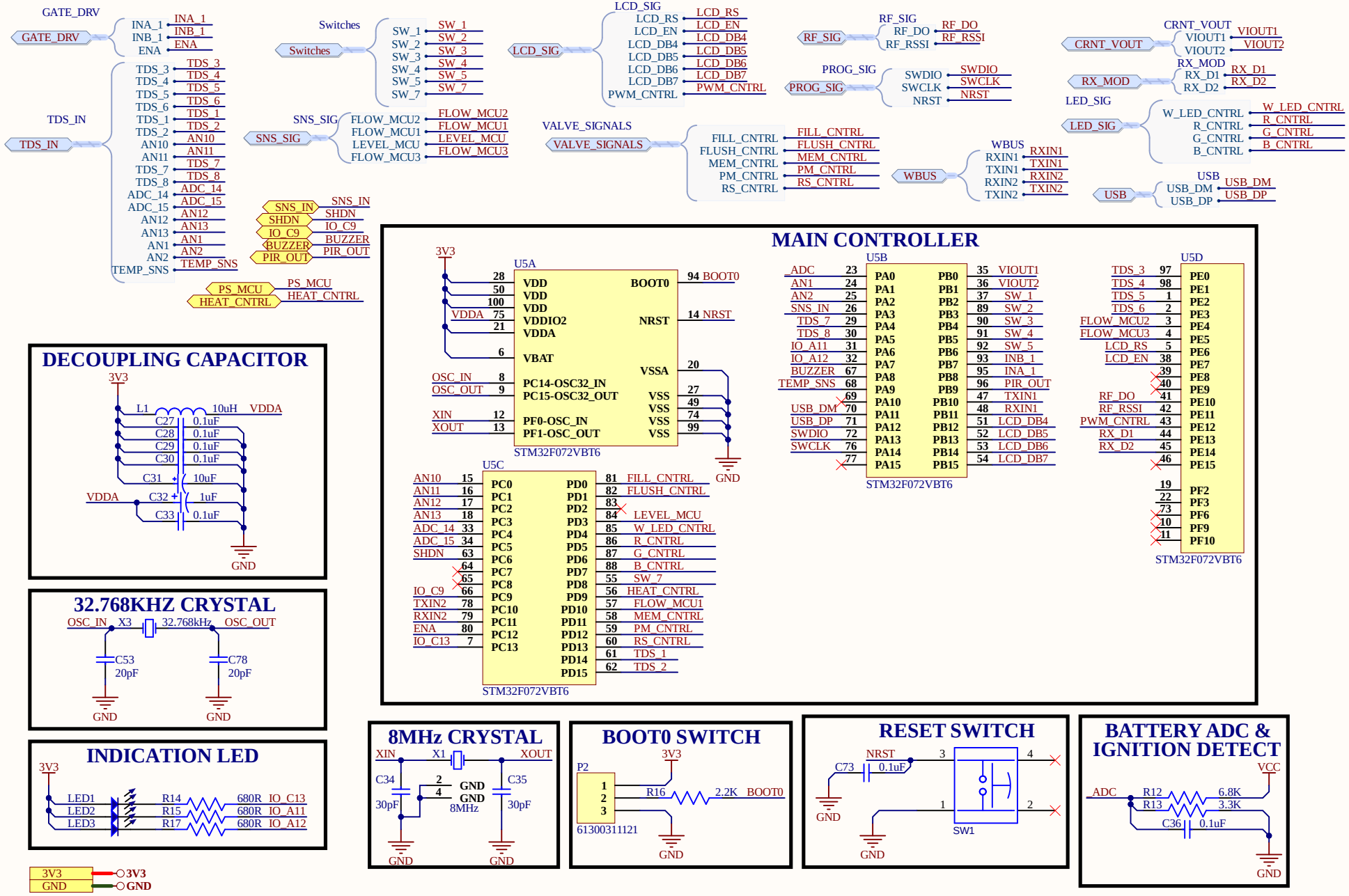


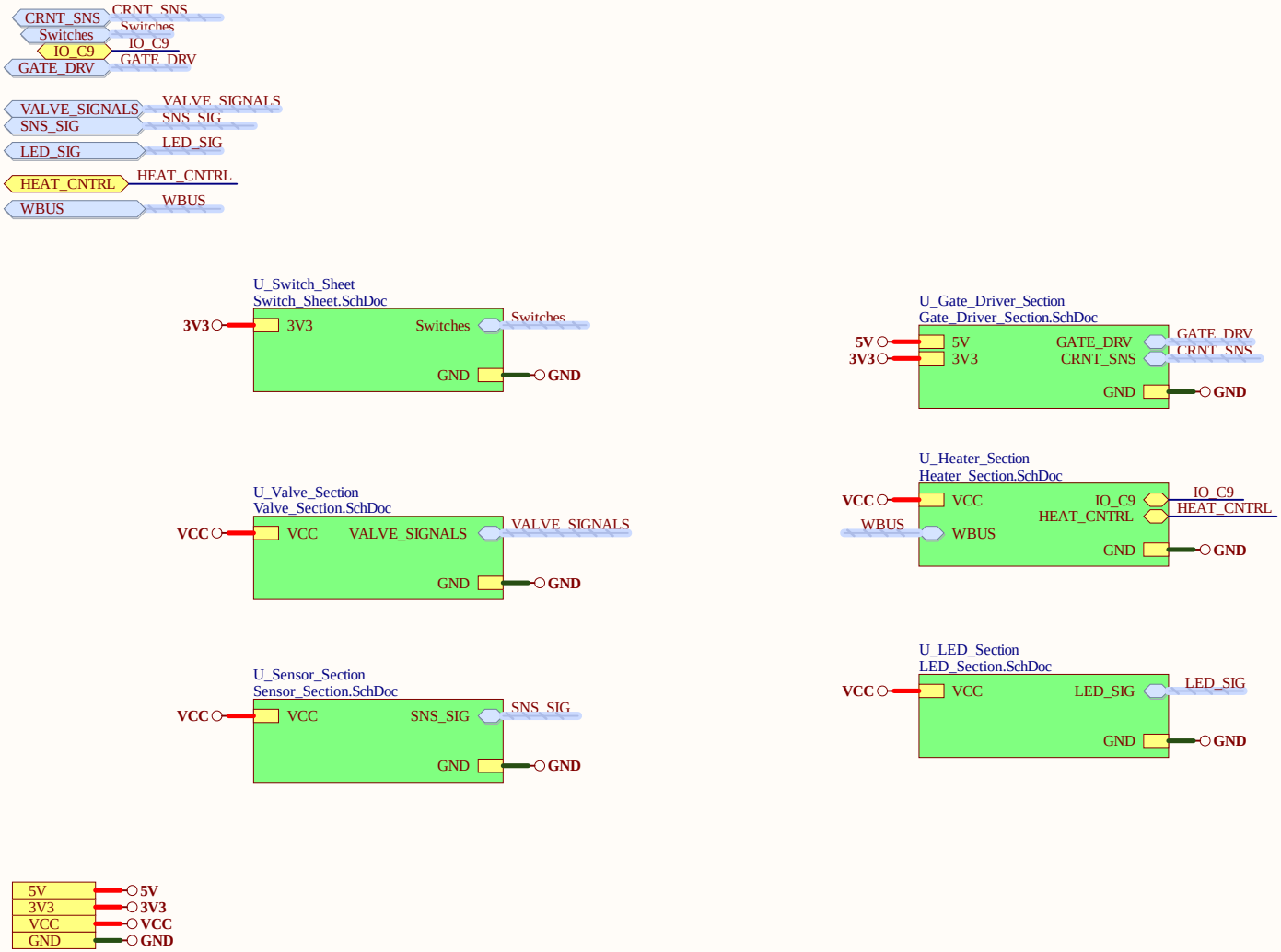
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

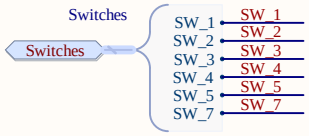
SUPPORTING CIRCUIT

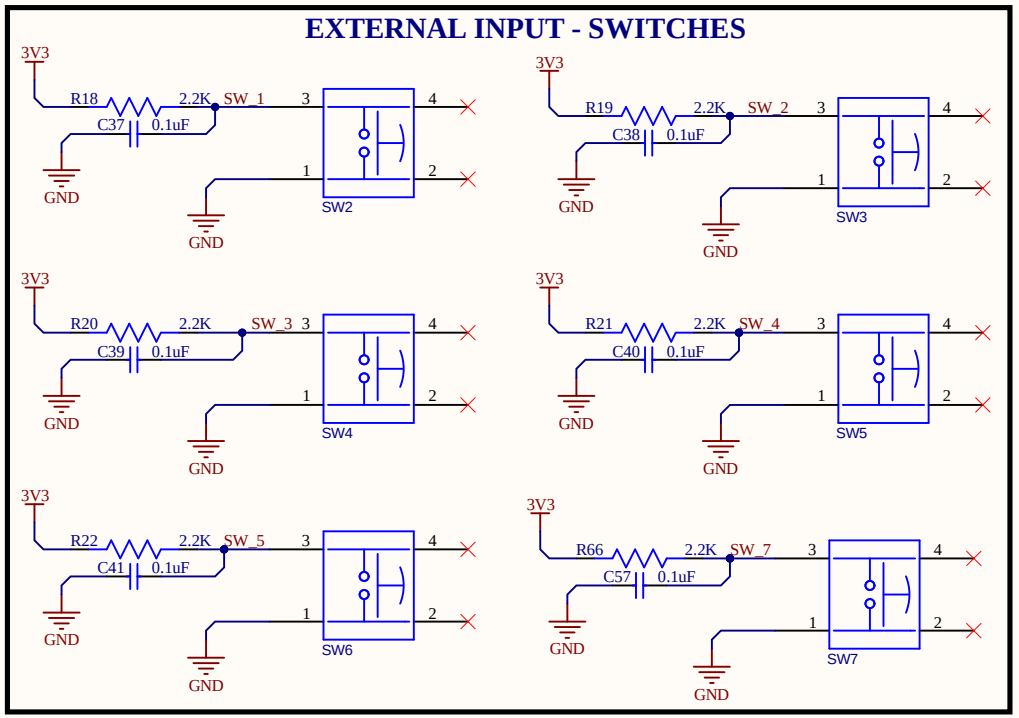


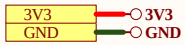



GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SNS_SIG
FLOW_MCU2
FLOW_MCU1
LEVEL_MCU
FLOW_MCU3
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
RF_SIG
RF_DO
RF_RSSI
PROG_SIG
SWDIO
SWCLK
NRST
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
USB
USB_DM
USB_DP
MAIN CONTROLLER
U5A
STM32F072VBT6
U5B
U5C
U5D
DECOUPLING CAPACITOR
3V3
L1
10uH
VDDA
C27
C28
C29
C30
C31
C32
C33
0.1uF
10uF
1uF
GND
32.768KHZ CRYSTAL
OSC_IN
X3
32.768kHz
OSC_OUT
C53
C78
20pF
INDICATION LED
LED1
LED2
LED3
R14
R15
R17
680R
IO_C13
IO_A11
IO_A12
8MHz CRYSTAL
XIN
X1
XOUT
C34
C35
30pF
8MHz
BOOT0 SWITCH
P2
R16
2.2K
BOOT0
61300311121
RESET SWITCH
NRST
C73
SW1
BATTERY ADC & IGNITION DETECT
VCC
ADC
R12
6.8K
R13
3.3K
C36


CRNT_SNS
Switches
IO_C9
GATE_DRV
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
WBUS
U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
Switches
GND
U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
3V3
GATE_DRV
CRNT_SNS
GND
U_Valve_Section
Valve_Section.SchDoc
VCC
VALVE_SIGNALS
GND
U_Heater_Section
Heater_Section.SchDoc
VCC
WBUS
IO_C9
HEAT_CNTRL
GND
U_Sensor_Section
Sensor_Section.SchDoc
VCC
SNS_SIG
GND
U_LED_Section
LED_Section.SchDoc
VCC
LED_SIG
GND
5V
3V3
VCC
GND


Switches
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7

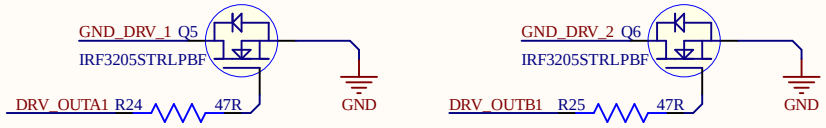

EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
SW2
R19
SW_2
C38
SW3
R20
SW_3
C39
SW4
R21
SW_4
C40
SW5
R22
SW_5
C41
SW6
R66
SW_7
C57
SW7


3V3
GND
3V3
GND

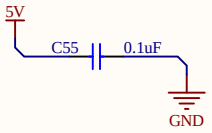
GATE_DRV
GATE_DRV
INA_1
INB_1
ENA
INA_1
INB_1
ENA

CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2

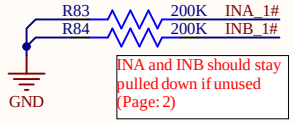
MOSFET DRIVERS



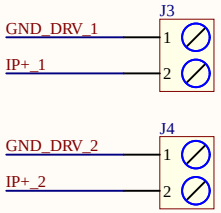
DECOUPLING CAPACITOR



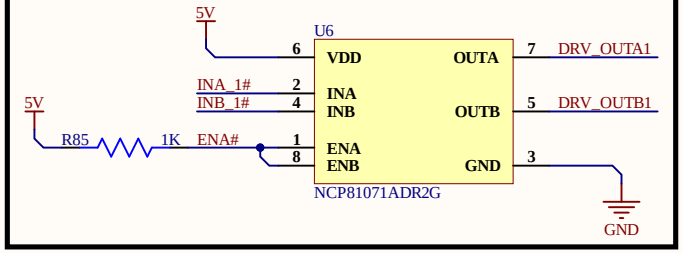
PULL DOWN RESISTOR



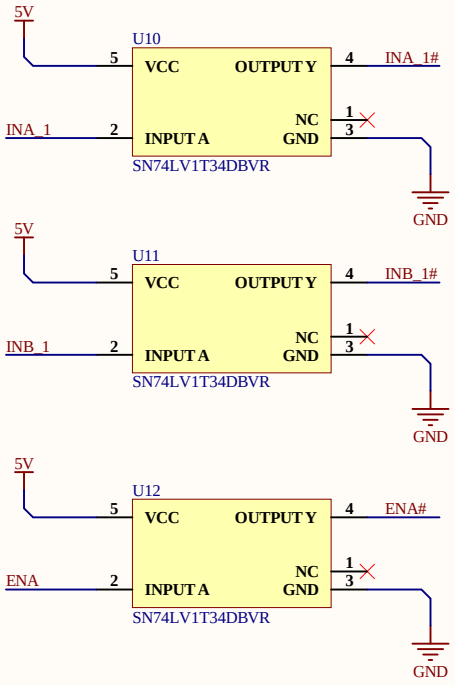
HEADERS FOR PUMP

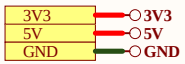


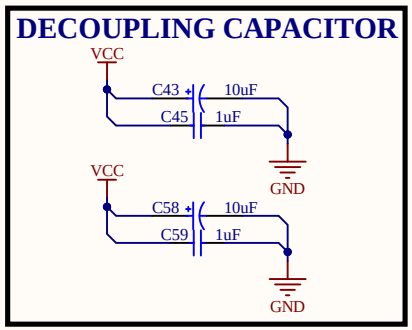
LOW SIDE PUMP DRIVER IC

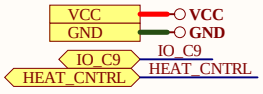


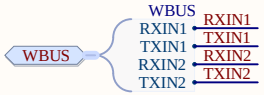
LEVEL TRANSLATOR

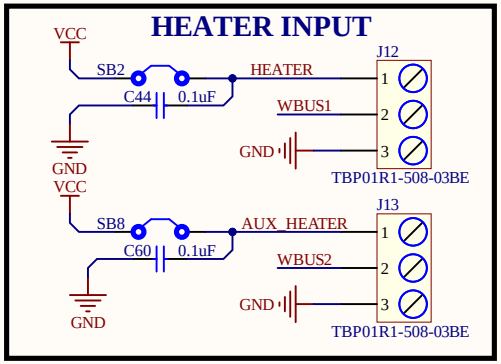


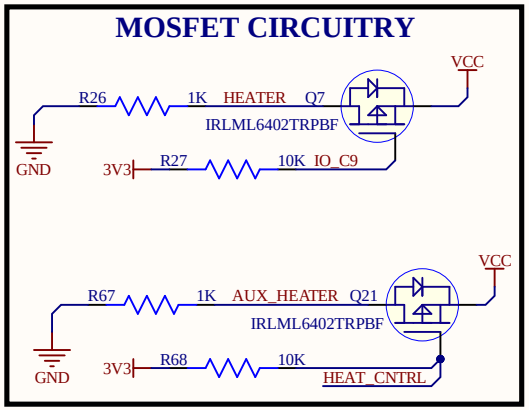


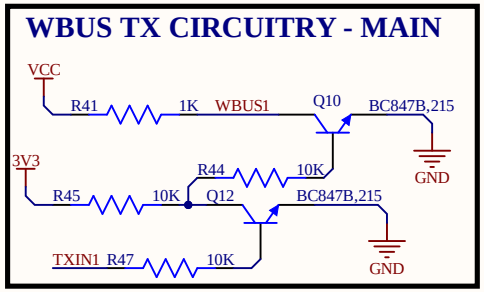

DECOUPLING CAPACITOR
VCC
C43
10uF
C45
1uF
GND
VCC
C58
10uF
C59
1uF
GND

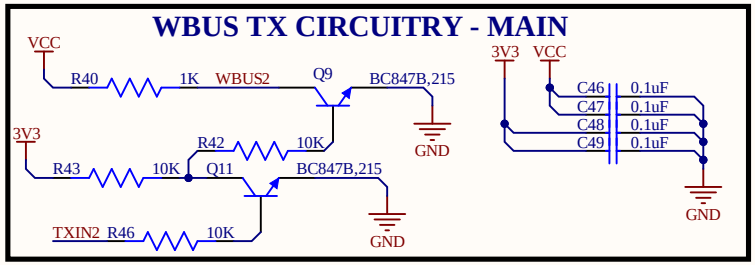

VCC
GND
IO_C9
HEAT_CNTRL
VCC
GND
IO_C9
HEAT_CNTRL

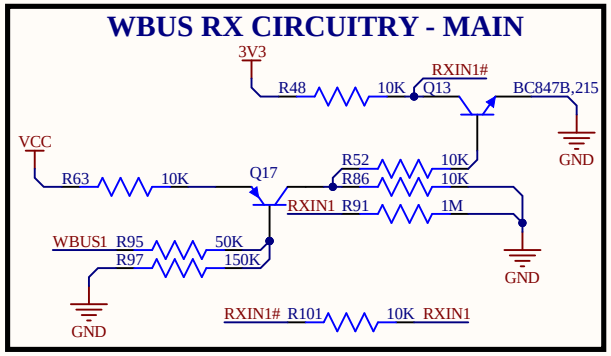

WBUS
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
RXIN1
TXIN1
RXIN2
TXIN2

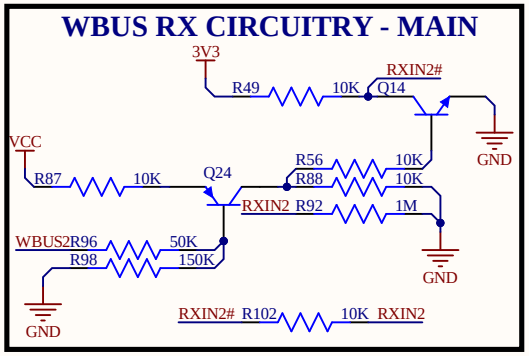

HEATER INPUT
VCC
SB2
C44
0.1uF
HEATER
J12
WBUS1
GND
TBP01R1-508-03BE
VCC
SB8
C60
0.1uF
AUX_HEATER
J13
WBUS2
GND
TBP01R1-508-03BE

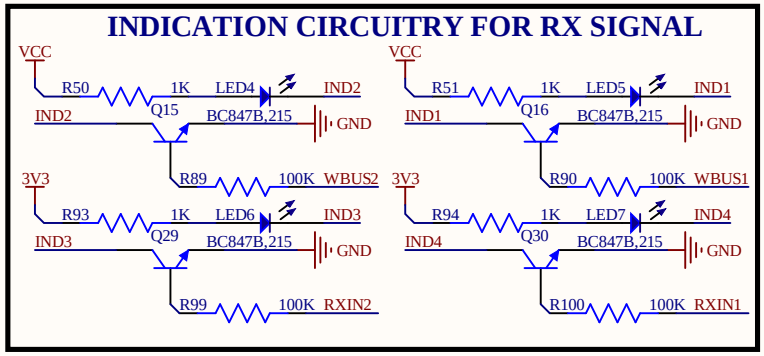

MOSFET CIRCUITRY
VCC
R26
1K
HEATER
Q7
IRLML6402TRPBF
GND
3V3
R27
10K
IO_C9
VCC
R67
1K
AUX_HEATER
Q21
IRLML6402TRPBF
GND
3V3
R68
10K
HEAT_CNTRL


WBUS TX CIRCUITRY - MAIN
VCC
R41
1K
WBUS1
Q10
BC847B,215
GND
3V3
R44
10K
R45
10K
Q12
BC847B,215
GND
TXIN1
R47
10K
GND


WBUS TX CIRCUITRY - MAIN
VCC
R40
1K
WBUS2
Q9
BC847B,215
GND
3V3
R42
10K
R43
10K
Q11
BC847B,215
GND
TXIN2
R46
10K
GND
3V3
VCC
C46
0.1uF
C47
0.1uF
C48
0.1uF
C49
0.1uF
GND


WBUS RX CIRCUITRY - MAIN
3V3
R48
10K
RXIN1#
Q13
BC847B,215
GND
VCC
R63
10K
Q17
R52
10K
R86
10K
RXIN1
R91
1M
GND
WBUS1
R95
50K
R97
150K
GND
RXIN1#
R101
10K
RXIN1


WBUS RX CIRCUITRY - MAIN
3V3
R49
10K
RXIN2#
Q14
GND
VCC
R87
10K
Q24
R56
10K
R88
10K
GND
RXIN2
R92
1M
WBUS2
R96
50K
R98
150K
GND
RXIN2#
R102
10K
RXIN2


INDICATION CIRCUITRY FOR RX SIGNAL
VCC
R50
1K
LED4
IND2
IND2
Q15
BC847B,215
GND
VCC
R51
1K
LED5
IND1
IND1
Q16
BC847B,215
GND
3V3
R89
100K
WBUS2
3V3
R90
100K
WBUS1
R93
1K
LED6
IND3
IND3
Q29
BC847B,215
GND
R94
1K
LED7
IND4
IND4
Q30
BC847B,215
GND
R99
100K
RXIN2
R100
100K
RXIN1

VALVE_SIGNALS

VALVE_SIGNALS

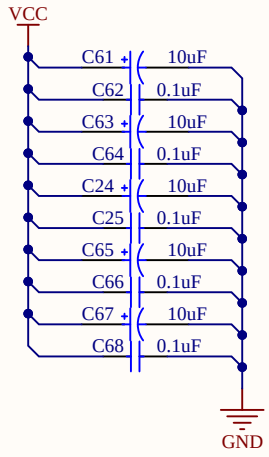
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL

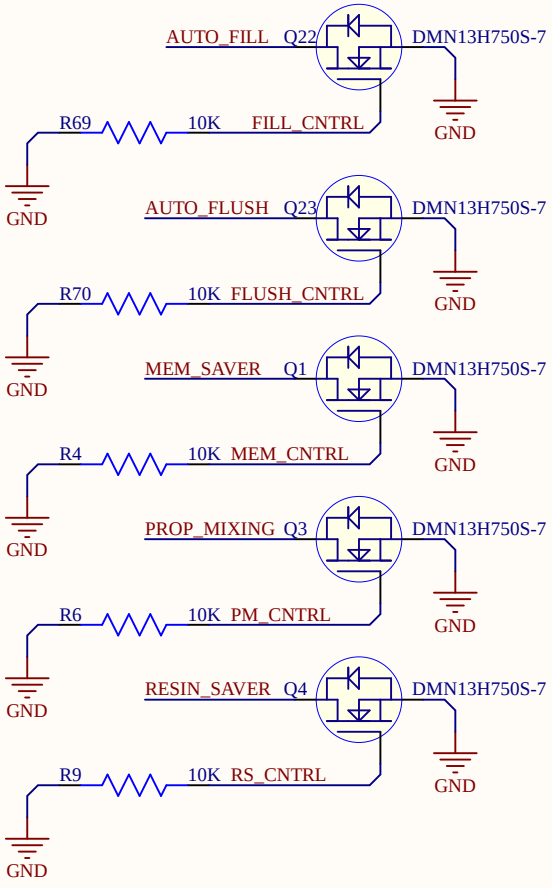
VCC
GND

VCC
GND

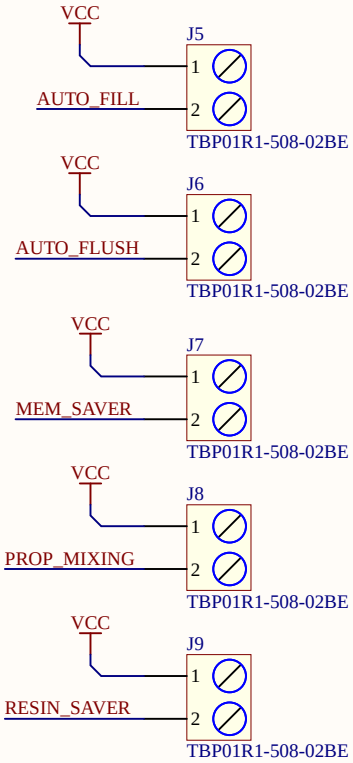
DECOUPLING CAPACITOR



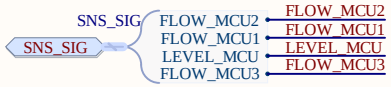
SOLENOID CONTROL CIRCUIT



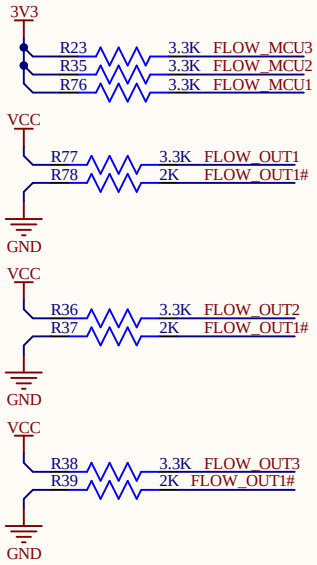
SOLENOID VALVE INPUTS



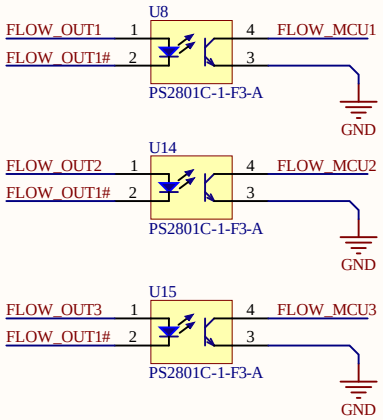
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Size A	Number	Revision
Date: 8/23/2022		Sheet of
File: C:\Users\..\Valve_Section.SchDoc		Drawn By:



PULL UP RESISTORS

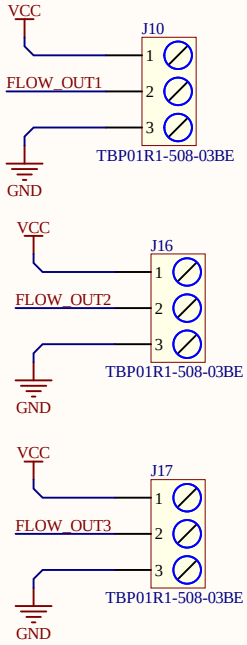


OPTO ISOLATION FOR FLOW METER

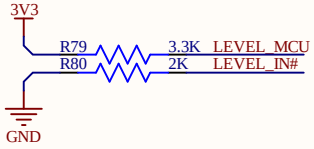


Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

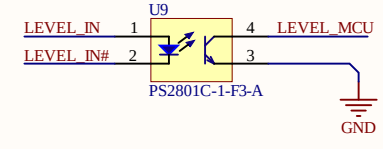
FLOW METER INPUT



SUPPORTING CIRCUIT

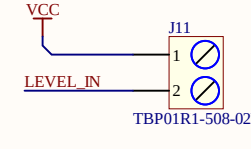


OPTO ISOLATION FOR FLOW METER



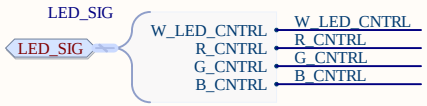
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

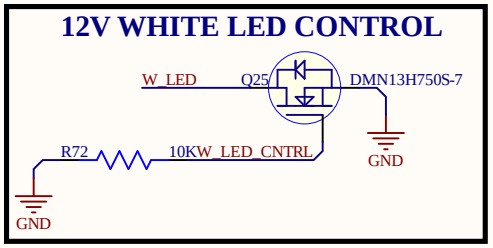
LEVEL SENSOR INPUT

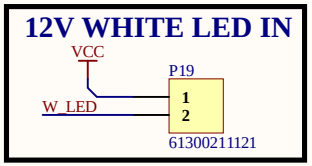


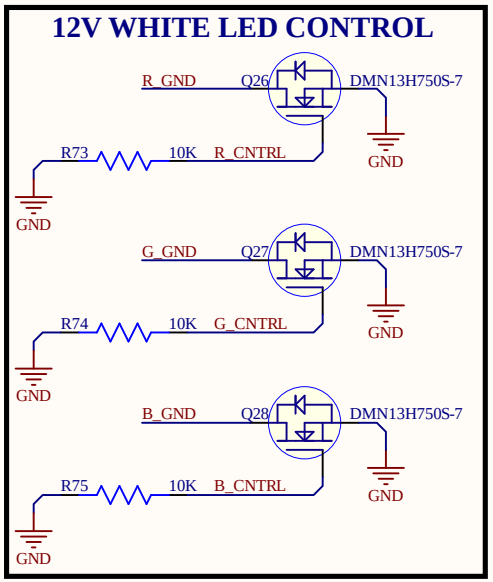


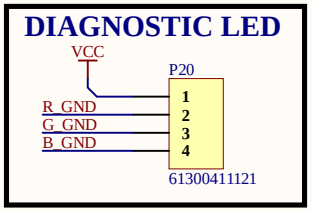
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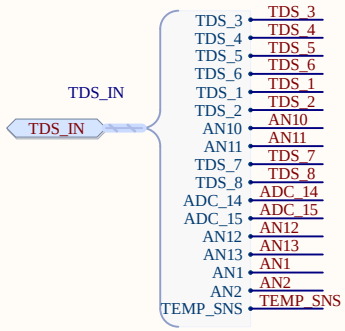




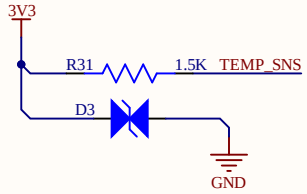




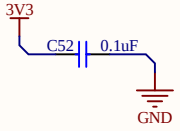
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Size: A4	Number	Revision
Date: 8/23/2022		Sheet of
File: C:\Users\..\LED_Section.SchDoc		Drawn By:



PULL UP RESISTORS

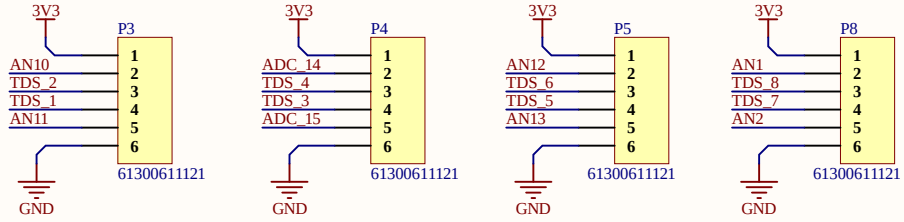


DECOUPLING CAP



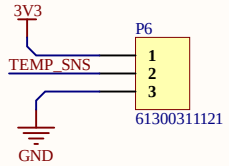


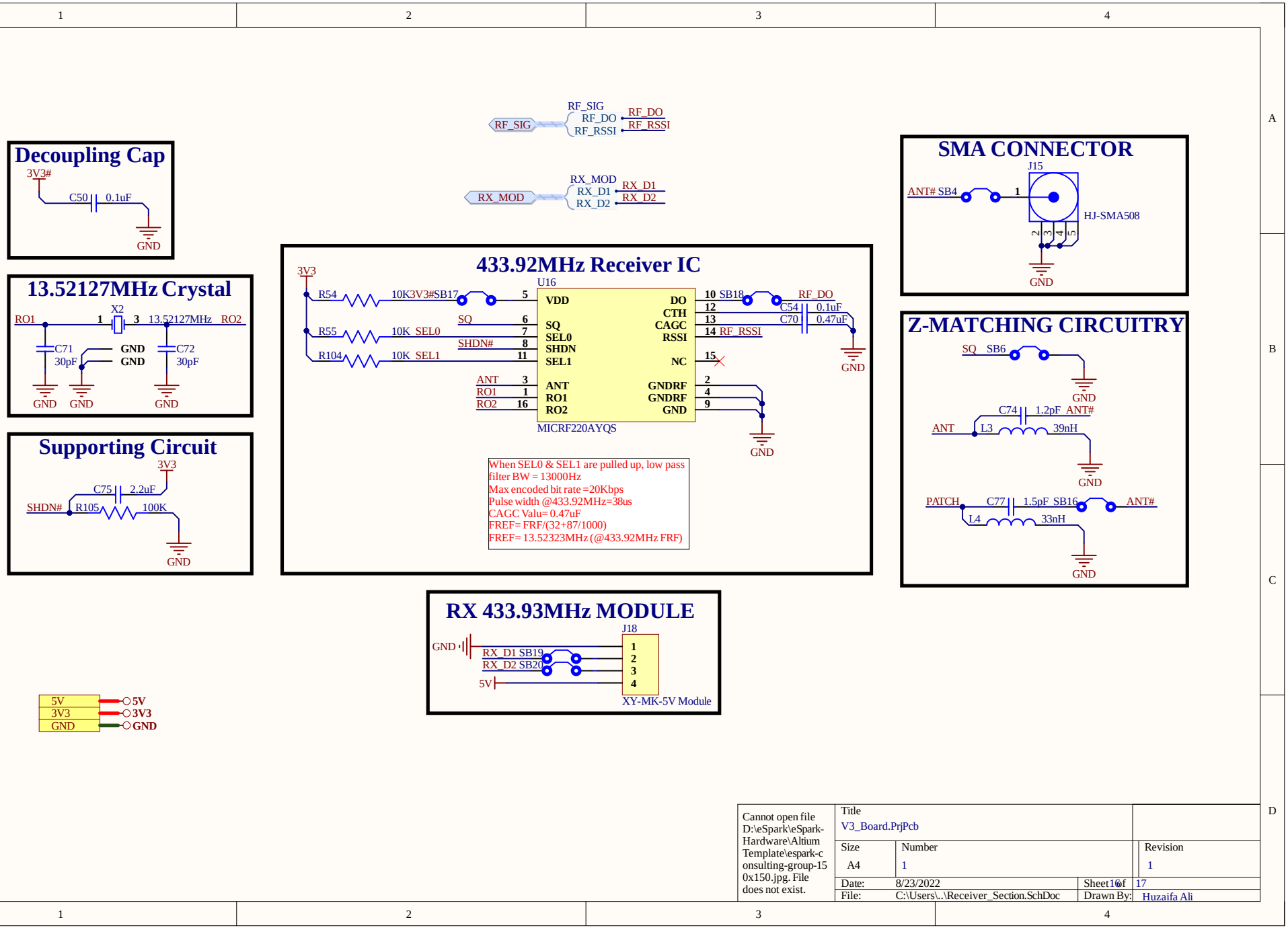
4 X TDS INPUTS



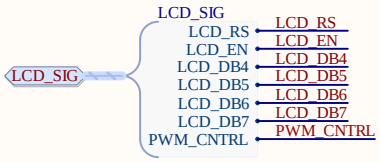
These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20



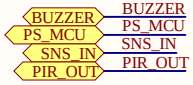


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: V3_Board.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/23/2022	Sheet 16 of 17	
	File: C:\Users\..\Receiver_Section.SchDoc	Drawn By: Huzaifa Ali	

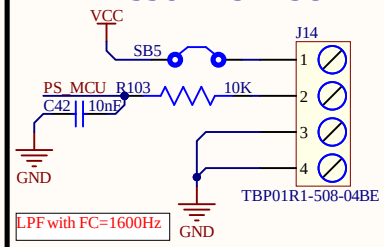




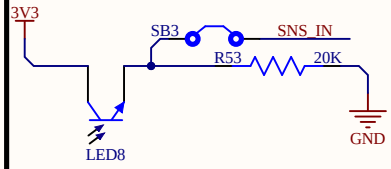




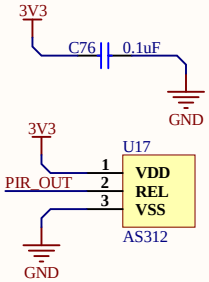
PRESSURE SENSOR



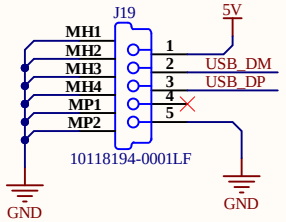
PHOTOTRANSISTOR



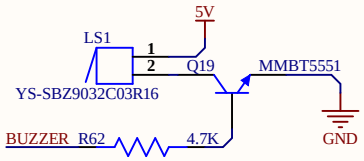
MOTION DETCTION



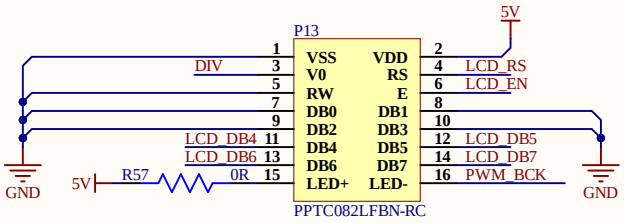
USB TYPE B CONNECTOR



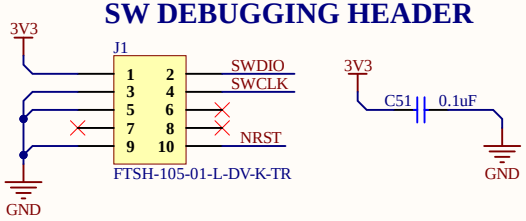
BUZZER CIRCUIT

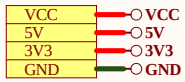


LCD CONNECTOR

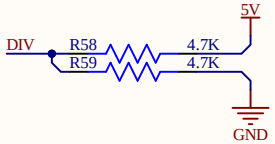


SW DEBUGGING HEADER





RESISTOR DIVIDER



PWM BACKLIGHT

